

```
%clear workspace and set figure style
clear all
close all

set(groot, 'defaultLineLineWidth', 2)
set(groot, 'DefaultAxesFontSize', 14)
set(groot, 'DefaultLegendFontSize', 14)
```

```
%global variables
```

```
global kb Rgas T0 Tref Patm atm2kPa
```

```
kb=8.6e-5
```

```
kb =
8.6000e-05
```

```
Rgas=8.3e-3
```

```
Rgas =
0.0083
```

```
T0=273.15
```

```
T0 =
273.1500
```

```
Tref=15
```

```
Tref =
15
```

```
Patm=0.209
```

```
Patm =
0.2090
```

```
atm2kPa=101.325
```

```
atm2kPa =
101.3250
```

```
%Andy's data
```

```
mpjoint=1; % 1=only use data with both MR+Pcrit for parameter estimation
```

```
%Pcrit in atm
```

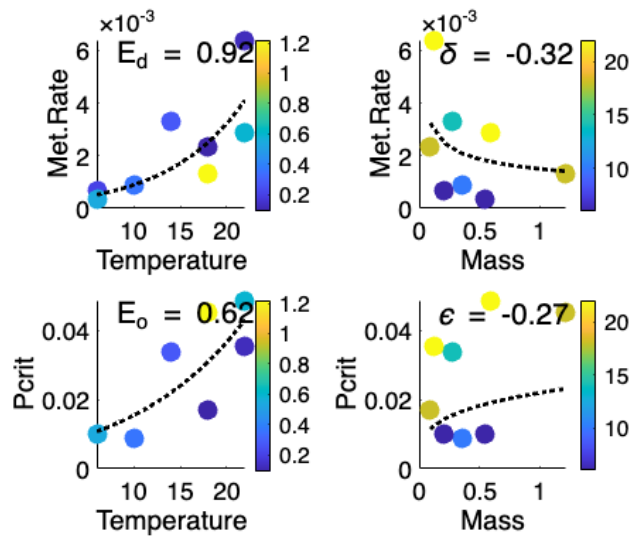
```
A=readtable('T.transversa_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the

VariableDescriptions property.

Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'T.transversa_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
    alphaD: [0.0017 5.2848e-04]
    delta: [-0.3202 0.1770]
    Ed: [0.9162 0.3534]
    r2m: 0.7840
    Ao: [41.6588 324.9439]
    epsilon: [-0.2680 0.0971]
    Eo: [0.6151 0.1558]
    r2p: 0.8670
```

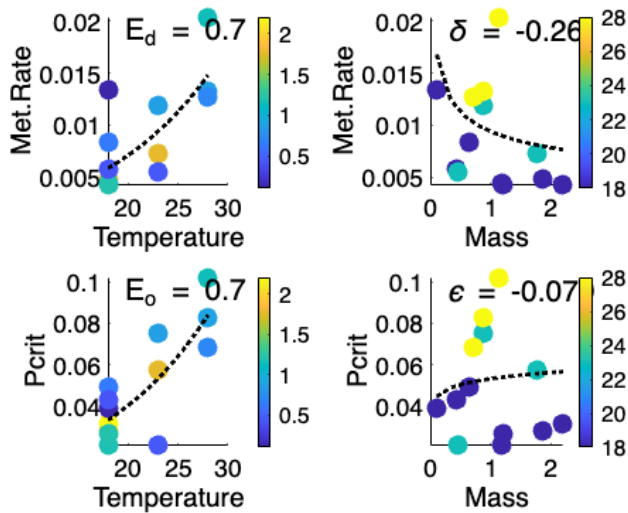
%Pcrit is atm

```
A=readtable('A.irradiations_FinalPhysData_atm_warmonly.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.

Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'A.irradiations_MIparam_atm_warmonly.jpg'
```



Par

```
Par = struct with fields:
    alphaD: [0.0044 0.0012]
    delta: [-0.2606 0.1518]
    Ed: [0.6983 0.1900]
    r2m: 0.6281
    Ao: [40.3312 177.5543]
    epsilon: [-0.0786 0.1673]
    Eo: [0.7002 0.1595]
    r2p: 0.6688
```

%Pcrit is atm

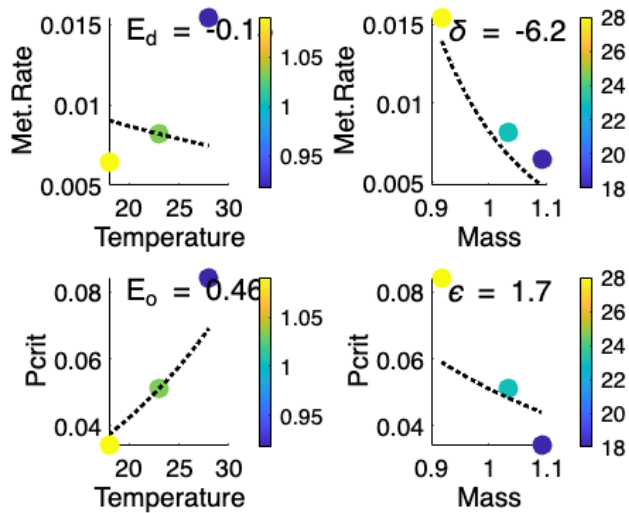
```
A=readtable('A.irradiations_TEST_meanPcrit_atm_warmonly.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'A.irradiations_TEST.jpg'
```



Par

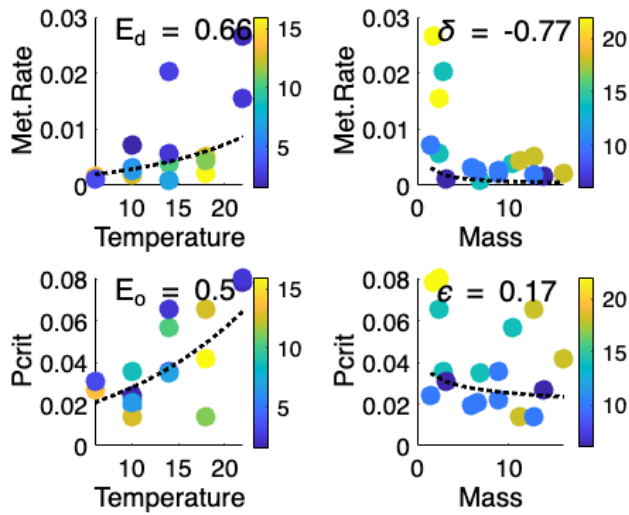
```
Par = struct with fields:
    alphaD: [0.0096 Inf]
    delta: [-6.1869 Inf]
    Ed: [-0.1452 Inf]
    r2m: 1
    Ao: [32.4603 0]
    epsilon: [1.6974 Inf]
    Eo: [0.4638 Inf]
    r2p: 1
```

%Pcrit in atm

```
A=readtable('M.franciscanus_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'M.franciscanus_MIparam_atm.jpg'
```



Par

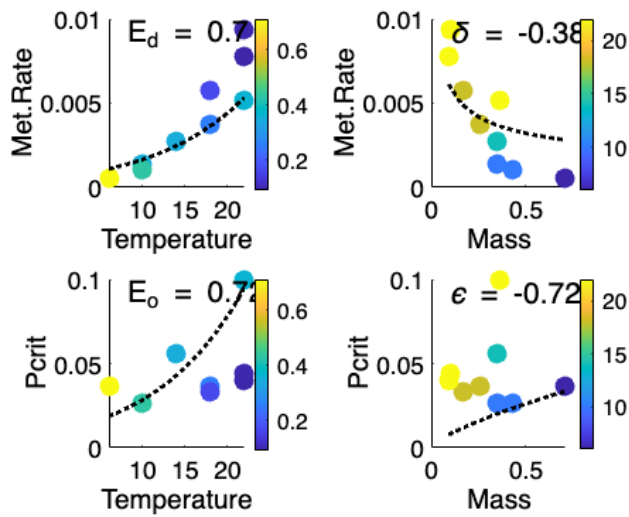
```
Par = struct with fields:
    alphaD: [0.0046 0.0013]
    delta: [-0.7685 0.2420]
    Ed: [0.6582 0.1815]
    r2m: 0.7626
    Ao: [24.9909 242.9356]
    epsilon: [0.1667 0.1136]
    Eo: [0.5029 0.1358]
    r2p: 0.5987
```

%Pcrit is in atm

```
A=readtable('C:\ligatum_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'C:\ligatum_MIparam_atm.jpg'
```



Par

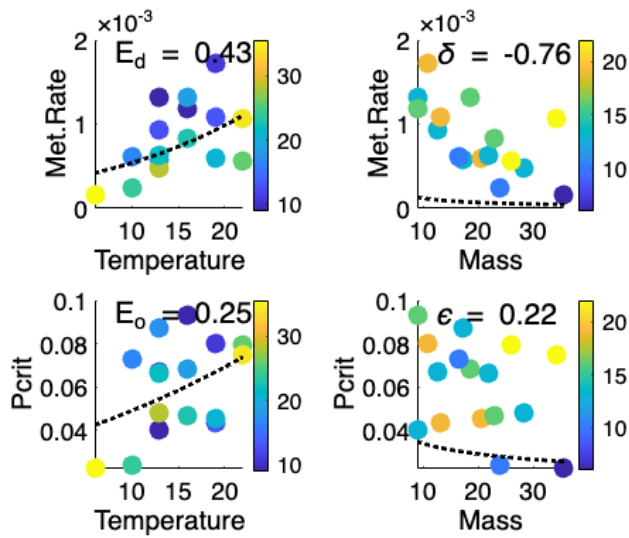
```
Par = struct with fields:
  alphaD: [0.0027 3.1965e-04]
  delta: [-0.3796 0.1082]
  Ed: [0.6993 0.1600]
  r2m: 0.9593
  Ao: [20.9720 261.8367]
  epsilon: [-0.7207 0.1367]
  Eo: [0.7198 0.1044]
  r2p: 0.8729
```

%Pcrit is in atm

```
A=readtable('K.kelletii_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'K.kelletii_MIparam_atm.jpg'
```



Par

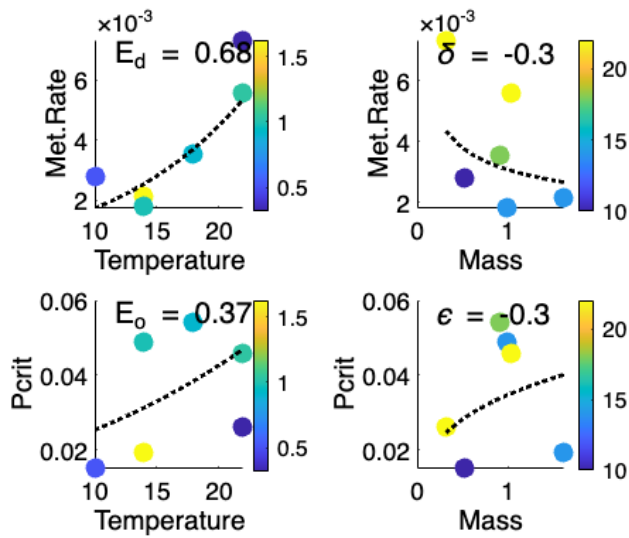
```
Par = struct with fields:
  alphaD: [7.3359e-04 7.4846e-05]
  delta: [-0.7632 0.1769]
  Ed: [0.4283 0.1584]
  r2m: 0.6855
  Ao: [17.1089 189.3258]
  epsilon: [0.2215 0.1912]
  Eo: [0.2453 0.1501]
  r2p: 0.2666
```

%Pcrit is in atm

```
A=readtable('L.hexactis_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'L.hexactis_MIparam_atm.jpg'
```



Par

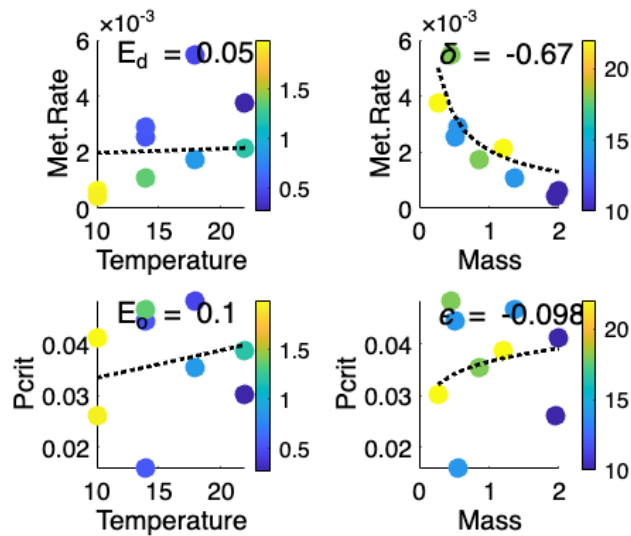
```
Par = struct with fields:
  alphaD: [0.0028 3.2041e-04]
  delta: [-0.3011 0.1137]
  Ed: [0.6776 0.1476]
  r2m: 0.9476
  Ao: [30.3771 116.7060]
  epsilon: [-0.3007 0.4381]
  Eo: [0.3707 0.3963]
  r2p: 0.3093
```

%Pcrit is in atm

```
A=readtable('G.septentrionalis_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'G.septentrionalis_MIparam_atm.jpg'
```

Par

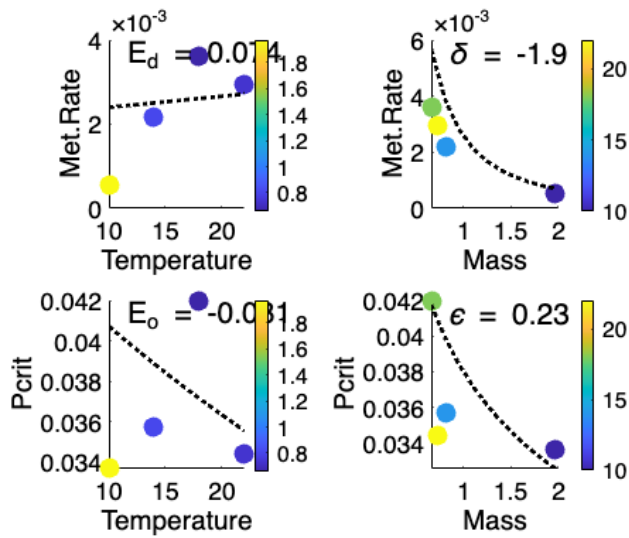
```
Par = struct with fields:
    alphaD: [0.0021 4.2461e-04]
    delta: [-0.6717 0.3364]
    Ed: [0.0518 0.3678]
    r2m: 0.6452
    Ao: [27.7085 243.6891]
    epsilon: [-0.0976 0.2072]
    Eo: [0.1045 0.2222]
    r2p: 0.0450
```

%Pcrit is in atm

```
A=readtable('G.septentrionalisTESTSE_FinalPhysData_atm.xls');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'G.septentrionalisTESTSE_MIparam_atm.jpg'
```



Par

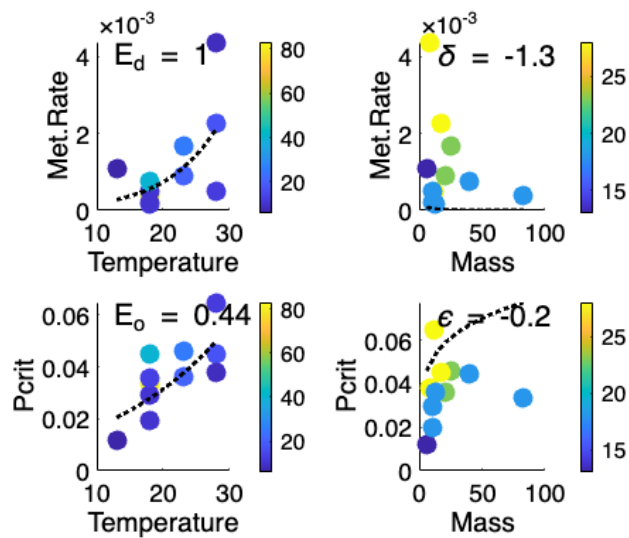
```
Par = struct with fields:
  alphaD: [0.0025 1.5451e-04]
  delta: [-1.9067 0.3706]
  Ed: [0.0739 0.1026]
  r2m: 0.9924
  Ao: [26.0321 280.4239]
  epsilon: [0.2274 0.2667]
  Eo: [-0.0815 0.1825]
  r2p: 0.4742
```

%Pcrit is in atm

```
A=readtable('S.sinistrum_FinalPhysData_atm_warmonly.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'S.sinistrum_MIparam_atm_warmonly.jpg'
```



Par

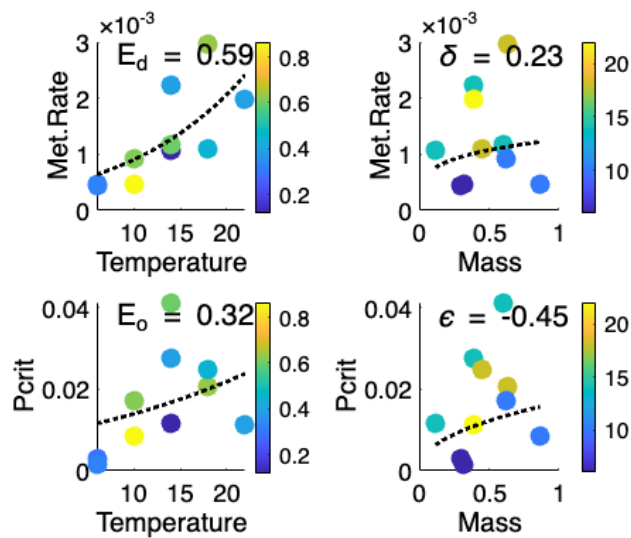
```
Par = struct with fields:
    alphaD: [3.7025e-04 2.7895e-04]
    delta: [-1.2903 0.7113]
    Ed: [1.0015 0.4373]
    r2m: 0.5384
    Ao: [43.0854 223.9272]
    epsilon: [-0.1969 0.1217]
    Eo: [0.4395 0.1349]
    r2p: 0.6505
```

%Pcrit is in atm

```
A=readtable('0.aculeata_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg '0.aculeata_MIparam_atm.jpg'
```



Par

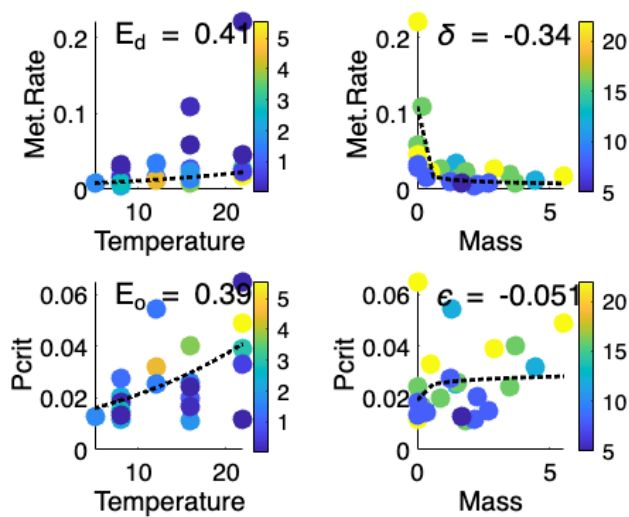
```
Par = struct with fields:
    alphaD: [0.0014 2.3286e-04]
    delta: [0.2335 0.3873]
    Ed: [0.5888 0.2478]
    r2m: 0.5180
    Ao: [56.9198 239.4213]
    epsilon: [-0.4524 0.5689]
    Eo: [0.3194 0.3309]
    r2p: 0.2490
```

%Pcrit is atm

```
A=readtable('S.droebachiensis_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'S.droebachiensis_MIparam_atm.jpg'
```



Par

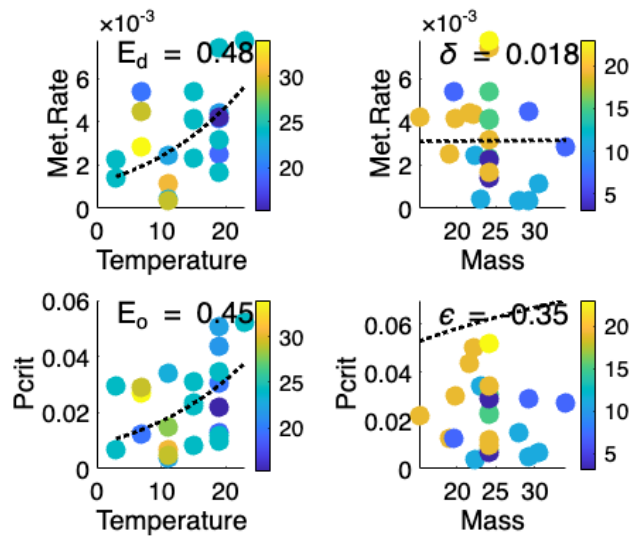
```
Par = struct with fields:
    alphaD: [0.0153 0.0045]
    delta: [-0.3413 0.0610]
    Ed: [0.4140 0.1989]
    r2m: 0.7961
    Ao: [35.3502 335.5352]
    epsilon: [-0.0506 0.0454]
    Eo: [0.3903 0.1363]
    r2p: 0.3137
```

%Pcrit is atm

```
A=readtable('F.serratissima_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'F.serratissima_MIparam_atm.jpg'
```



Par

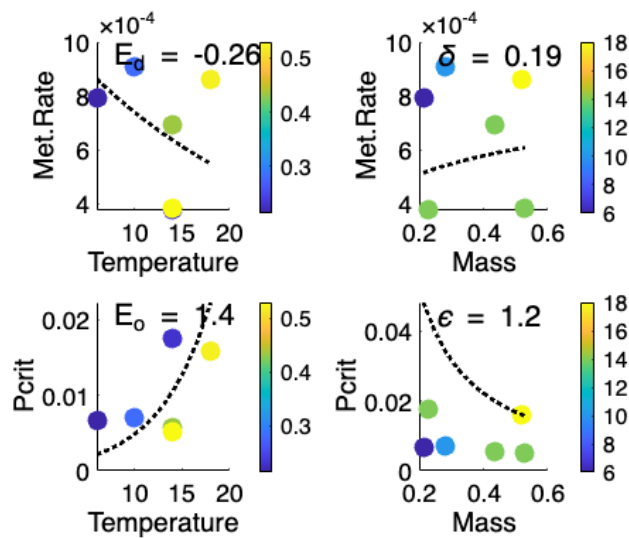
```
Par = struct with fields:
  alphaD: [0.0034 4.2097e-04]
  delta: [0.0181 0.7653]
  Ed: [0.4752 0.1984]
  r2m: 0.2576
  Ao: [42.9549 331.5227]
  epsilon: [-0.3476 0.8764]
  Eo: [0.4456 0.2108]
  r2p: 0.1817
```

```
%Pcrit is atm
```

```
A=readtable('T.septentrionalis_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'T.septentrionalis_MIparam_atm.jpg'
```



Par

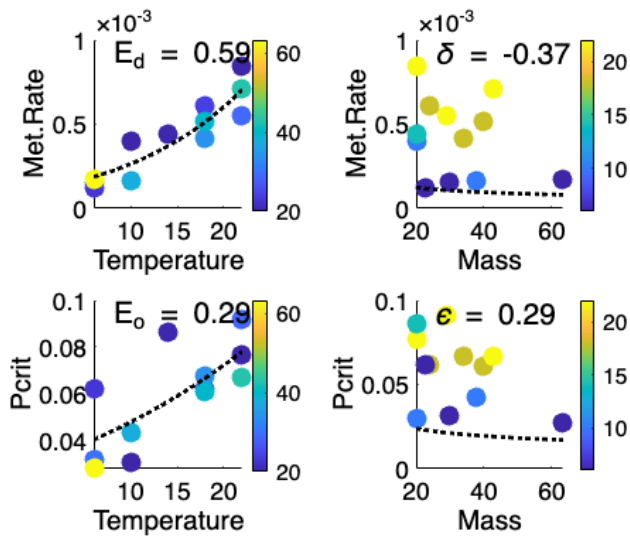
```
Par = struct with fields:
  alphaD: [6.1157e-04 1.5591e-04]
  delta: [0.1855 0.7654]
  Ed: [-0.2646 0.5103]
  r2m: 0.0990
  Ao: [78.9083 806.9493]
  epsilon: [1.2376 0.3276]
  Eo: [1.3538 0.3450]
  r2p: 0.8759
```

```
%Pcrit is atm
```

```
A=readtable('M.arenaria_FinalPhysData_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'M.arenaria_MIparam_atm.jpg'
```



Par

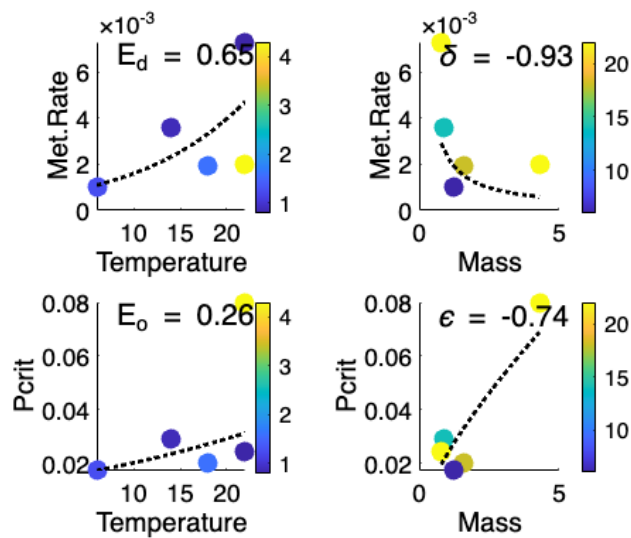
```
Par = struct with fields:
  alphaD: [4.0120e-04 3.0070e-05]
  delta: [-0.3658 0.1809]
  Ed: [0.5870 0.0913]
  r2m: 0.8864
  Ao: [17.0015 226.8798]
  epsilon: [0.2882 0.2251]
  Eo: [0.2950 0.0913]
  r2p: 0.6248
```

```
%Pcrit in atm
```

```
A=readtable('Henricia_FinalPhysData_atm.xls');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'Henricia_MIparam_atm.jpg'
```

Par

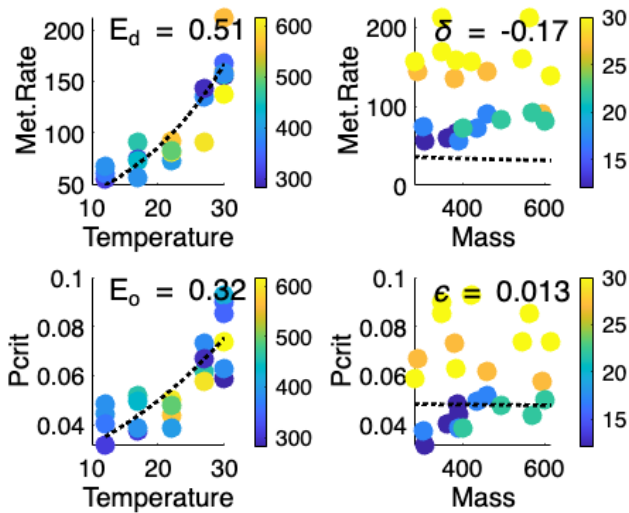
```
Par = struct with fields:
  alphaD: [0.0025 3.8160e-04]
  delta: [-0.9347 0.2324]
  Ed: [0.6479 0.1702]
  r2m: 0.9635
  Ao: [41.0366 154.4991]
  epsilon: [-0.7382 0.2876]
  Eo: [0.2622 0.3491]
  r2p: 0.8931
```

```
% Phys Lit Review Deutsch 2020%
% SMR and Mass = 1, don't use %
```

```
%C. striata
%Pcrit in atm
A=readtable('c.striata_formatlab.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.weight_gms,A.pcrit,A.RMR,mpjoint);
print -djpeg 'c.striata_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
    alphaD: [60.6703 6.3904]
    delta: [-0.1668 0.1382]
    Ed: [0.5074 0.0599]
    r2m: 0.8352
    Ao: [24.9389 311.9591]
    epsilon: [0.0128 0.1385]
    Eo: [0.3163 0.0483]
    r2p: 0.7102
```

```
%A.smithi
%Pcrit is atm
A=readtable('Deutsch2020PubData_Acanthephyra_smithi_Pcrit.atm.csv');
```

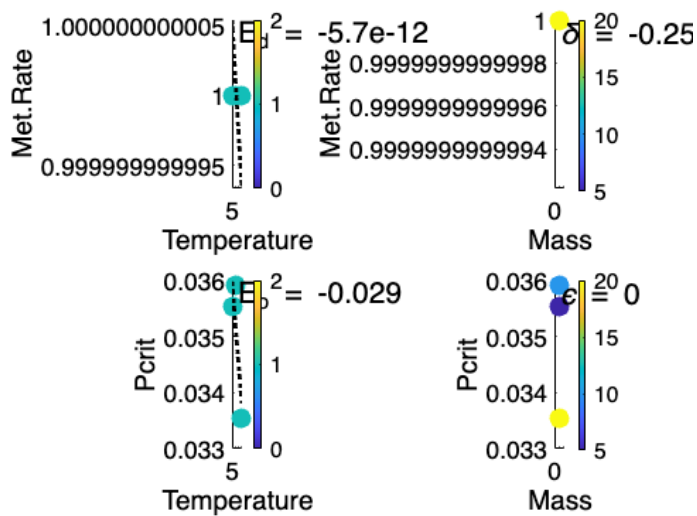
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

```
Warning: Rank deficient, rank = 2, tol = 2.316904e-15.
Warning: Rank deficient, rank = 2, tol = 3.126841e-15.
Warning: Rank deficient, rank = 2, tol = 2.407750e-15.
Warning: Rank deficient, rank = 2, tol = 2.341619e-15.
Warning: Rank deficient, rank = 2, tol = 2.340791e-15.
Warning: Rank deficient, rank = 1, tol = 2.307555e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will
not be able to compute confidence or prediction intervals, and you should use caution in making
predictions.
Warning: Rank deficient, rank = 2, tol = 3.150550e-15.
Warning: Rank deficient, rank = 2, tol = 2.432657e-15.
Warning: Rank deficient, rank = 2, tol = 2.345573e-15.
Warning: Rank deficient, rank = 2, tol = 2.342375e-15.
Warning: Rank deficient, rank = 2, tol = 2.342385e-15.
```

Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'A.smithi_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 3.5448e-12]
  delta: [-0.2500 0]
  Ed: [-5.6510e-12 0]
  r2m: -Inf
  Ao: [28.9930 1.7099e+03]
  epsilon: [0 0]
  Eo: [-0.0291 0.0165]
  r2p: 0.7611
```

```
%G.zoea
%Pcrit is atm
A=readtable('Deutsch2020PubData_Gnathopausia_zoea_Pcrit_atm.csv');
```

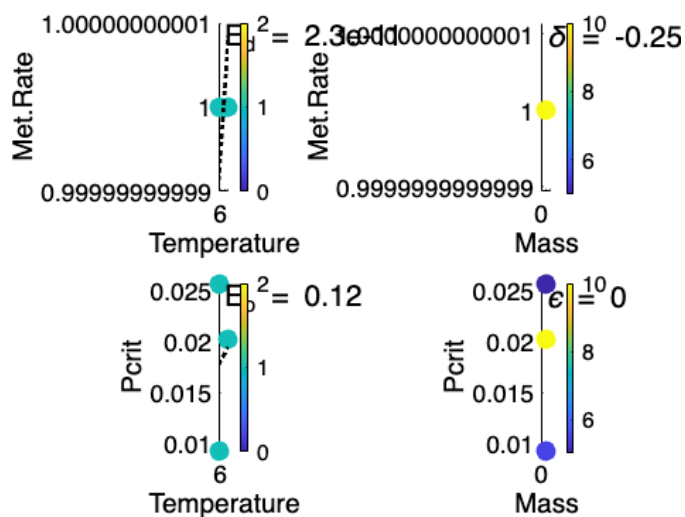
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

```
Warning: Rank deficient, rank = 2, tol = 1.224793e-15.
Warning: Rank deficient, rank = 2, tol = 1.278199e-15.
Warning: Rank deficient, rank = 2, tol = 4.107324e-15.
Warning: Rank deficient, rank = 2, tol = 2.983610e-15.
Warning: Rank deficient, rank = 2, tol = 2.830981e-15.
Warning: Rank deficient, rank = 2, tol = 2.827952e-15.
Warning: Rank deficient, rank = 2, tol = 2.827648e-15.
```

Warning: Rank deficient, rank = 2, tol = 3.063057e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: Rank deficient, rank = 2, tol = 3.779046e-15.
Warning: Rank deficient, rank = 2, tol = 2.684018e-15.
Warning: Rank deficient, rank = 2, tol = 2.090668e-15.
Warning: Rank deficient, rank = 2, tol = 2.004621e-15.
Warning: Rank deficient, rank = 2, tol = 1.999442e-15.
Warning: Rank deficient, rank = 2, tol = 1.999104e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'G.zoea_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 2.1606e-11]
  delta: [-0.2500 0]
  Ed: [2.3344e-11 1.6277e-11]
  r2m: -Inf
  Ao: [47.1035 36.0255]
  epsilon: [0 0]
  Eo: [0.1231 1.0887]
  r2p: 0.0115
```

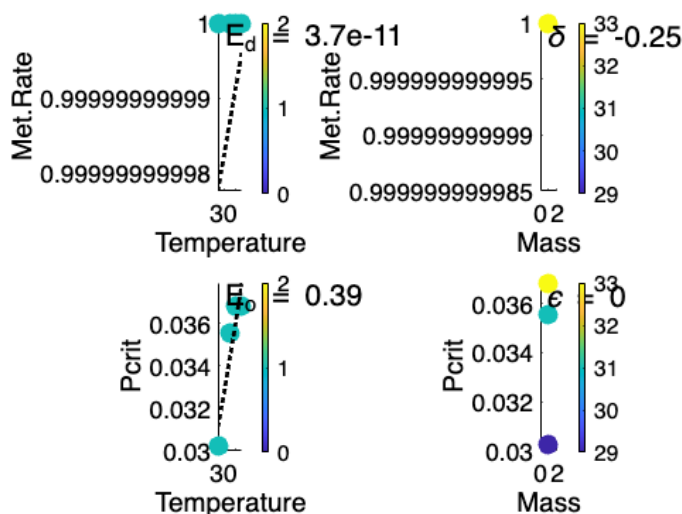
```
%G.erythrospilus
%Pcrit is atm
A=readtable('Deutsch2020PubData_Gobiodon_erythrospilus_Pcrit.atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 2.828513e-14.
Warning: Rank deficient, rank = 2, tol = 1.194185e-14.
Warning: Rank deficient, rank = 2, tol = 7.686829e-15.
Warning: Rank deficient, rank = 2, tol = 6.474389e-15.
Warning: Rank deficient, rank = 2, tol = 6.686011e-15.
Warning: Rank deficient, rank = 2, tol = 6.718869e-15.
Warning: Rank deficient, rank = 2, tol = 6.271587e-15.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.
Warning: Rank deficient, rank = 2, tol = 1.327730e-15.
Warning: Rank deficient, rank = 2, tol = 6.681645e-15.
Warning: Rank deficient, rank = 2, tol = 6.689806e-15.
Warning: Rank deficient, rank = 2, tol = 7.274199e-15.
Warning: Rank deficient, rank = 2, tol = 7.278575e-15.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.

```
print -djpeg 'G.erythropilus_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 7.3961e-11]
  delta: [-0.2500 0]
  Ed: [3.6772e-11 3.6660e-11]
  r2m: -Inf
  Ao: [66.9720 279.3440]
  epsilon: [0 0]
  Eo: [0.3927 0.1096]
  r2p: 0.8723
```

```
%G.histrio
%Pcrit is atm
```

```
A=readtable('Deutsch2020PubData_Gobiodon_histrio_Pcrit.atm.csv');
```

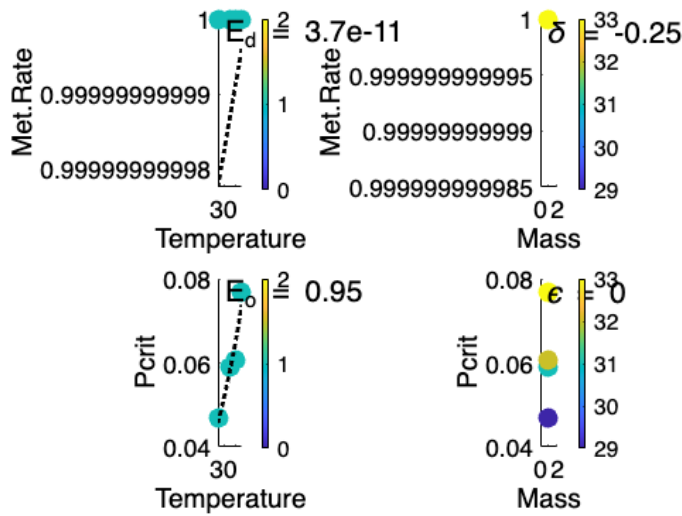
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 2.828513e-14.
Warning: Rank deficient, rank = 2, tol = 1.194185e-14.
Warning: Rank deficient, rank = 2, tol = 7.686829e-15.
Warning: Rank deficient, rank = 2, tol = 6.474389e-15.
Warning: Rank deficient, rank = 2, tol = 6.686011e-15.
Warning: Rank deficient, rank = 2, tol = 6.718869e-15.
Warning: Rank deficient, rank = 2, tol = 6.271587e-15.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.
Warning: Rank deficient, rank = 2, tol = 1.327730e-15.
Warning: Rank deficient, rank = 2, tol = 1.385624e-15.
Warning: Rank deficient, rank = 2, tol = 3.262975e-15.
Warning: Rank deficient, rank = 2, tol = 2.869441e-15.
Warning: Rank deficient, rank = 2, tol = 2.994560e-15.
Warning: Rank deficient, rank = 2, tol = 3.137813e-15.
Warning: Rank deficient, rank = 2, tol = 3.226103e-15.
Warning: Rank deficient, rank = 2, tol = 3.366774e-15.
Warning: Rank deficient, rank = 2, tol = 3.606003e-15.
Warning: Rank deficient, rank = 2, tol = 3.690701e-15.
Warning: Rank deficient, rank = 2, tol = 3.851629e-15.
Warning: Rank deficient, rank = 2, tol = 4.103176e-15.
Warning: Rank deficient, rank = 2, tol = 4.178063e-15.
Warning: Rank deficient, rank = 2, tol = 4.360243e-15.
Warning: Rank deficient, rank = 2, tol = 4.622430e-15.
Warning: Rank deficient, rank = 2, tol = 4.685052e-15.
Warning: Rank deficient, rank = 2, tol = 4.889338e-15.
Warning: Rank deficient, rank = 2, tol = 5.160566e-15.
Warning: Rank deficient, rank = 2, tol = 5.208603e-15.
Warning: Rank deficient, rank = 2, tol = 5.435718e-15.
Warning: Rank deficient, rank = 2, tol = 5.714403e-15.
Warning: Rank deficient, rank = 2, tol = 5.745701e-15.
Warning: Rank deficient, rank = 2, tol = 5.996236e-15.
Warning: Rank deficient, rank = 2, tol = 6.280838e-15.
Warning: Rank deficient, rank = 2, tol = 6.293423e-15.
Warning: Rank deficient, rank = 2, tol = 8.698447e-15.
Warning: Rank deficient, rank = 2, tol = 8.737463e-15.
Warning: Rank deficient, rank = 2, tol = 1.166642e-14.
Warning: Rank deficient, rank = 2, tol = 1.401466e-14.
Warning: Rank deficient, rank = 2, tol = 1.407752e-14.
Warning: Rank deficient, rank = 2, tol = 1.613717e-14.
Warning: Rank deficient, rank = 2, tol = 1.778043e-14.
Warning: Rank deficient, rank = 2, tol = 1.786018e-14.
Warning: Rank deficient, rank = 2, tol = 1.928452e-14.
Warning: Rank deficient, rank = 2, tol = 2.035873e-14.
Warning: Rank deficient, rank = 2, tol = 2.360051e-14.
Warning: Rank deficient, rank = 2, tol = 2.494370e-14.
Warning: Rank deficient, rank = 2, tol = 2.501353e-14.
Warning: Rank deficient, rank = 2, tol = 2.501572e-14.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to

numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.

```
print -djpeg 'G.histrio_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 7.3961e-11]
  delta: [-0.2500 0]
  Ed: [3.6772e-11 3.6660e-11]
  r2m: -Inf
  Ao: [130.4210 292.8402]
  epsilon: [0 0]
  Eo: [0.9547 0.2007]
  r2p: 0.9253
```

```
%M.pammelas
%Pcrit is atm
A=readtable('Deutsch2020PubData_Melanostigma_pammelas_Pcrit.atm.csv');
```

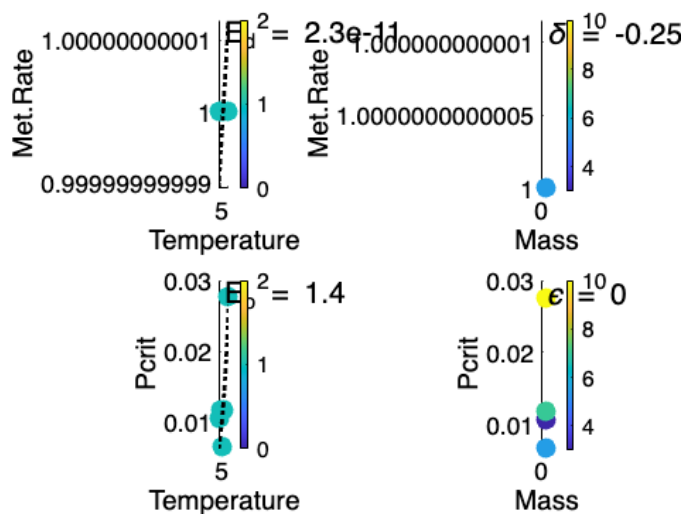
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

```
Warning: Rank deficient, rank = 2, tol = 2.148430e-15.
Warning: Rank deficient, rank = 2, tol = 2.242110e-15.
Warning: Rank deficient, rank = 2, tol = 8.250630e-15.
Warning: Rank deficient, rank = 2, tol = 5.700492e-15.
Warning: Rank deficient, rank = 2, tol = 5.313255e-15.
Warning: Rank deficient, rank = 2, tol = 5.302064e-15.
Warning: Rank deficient, rank = 2, tol = 5.302049e-15.
Warning: Rank deficient, rank = 2, tol = 5.897446e-15.
Warning: Rank deficient, rank = 2, tol = 5.897472e-15.
Warning: Rank deficient, rank = 2, tol = 5.897738e-15.
Warning: Rank deficient, rank = 2, tol = 5.900391e-15.
```

Warning: Rank deficient, rank = 2, tol = 5.926857e-15.
Warning: Rank deficient, rank = 2, tol = 6.185290e-15.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.
Warning: Rank deficient, rank = 2, tol = 6.808935e-15.
Warning: Rank deficient, rank = 2, tol = 2.649545e-15.
Warning: Rank deficient, rank = 2, tol = 9.902568e-16.
Warning: Rank deficient, rank = 2, tol = 8.260887e-16.
Warning: Rank deficient, rank = 2, tol = 8.451919e-16.
Warning: Rank deficient, rank = 2, tol = 8.466286e-16.
Warning: Rank deficient, rank = 2, tol = 8.468195e-16.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.

```
print -djpeg 'M.pammelas_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 2.0916e-11]
  delta: [-0.2500 0]
  Ed: [2.2666e-11 1.4087e-11]
  r2m: -Inf
  Ao: [14.0536 38.0960]
  epsilon: [0 0]
  Eo: [1.3864 0.3683]
  r2p: 0.8192
```

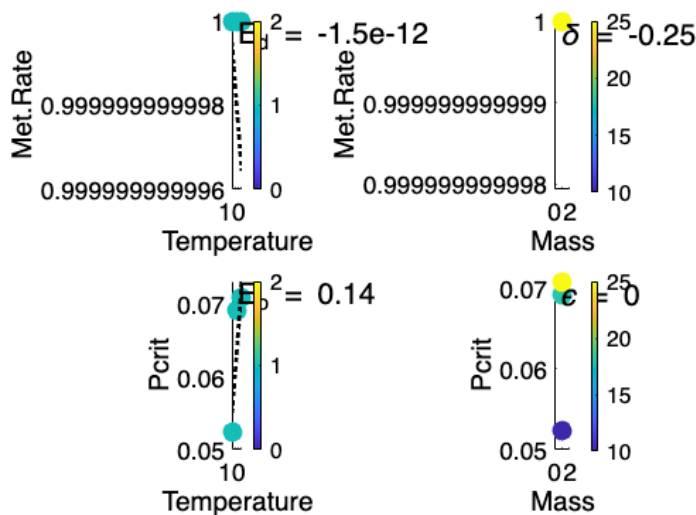
```
%Mytilu.spp
%Pcrit is atm
A=readtable('Deutsch2020PubData_Mytilus_spp_Pcrit_atm.csv');
```


Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 4.468923e-15.
Warning: Rank deficient, rank = 2, tol = 2.624281e-15.
Warning: Rank deficient, rank = 2, tol = 2.322380e-15.
Warning: Rank deficient, rank = 2, tol = 2.307774e-15.
Warning: Rank deficient, rank = 2, tol = 2.307556e-15.
Warning: Rank deficient, rank = 2, tol = 2.329442e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: Rank deficient, rank = 2, tol = 2.250037e-15.
Warning: Rank deficient, rank = 2, tol = 2.551039e-15.
Warning: Rank deficient, rank = 2, tol = 2.461853e-15.
Warning: Rank deficient, rank = 2, tol = 2.454696e-15.
Warning: Rank deficient, rank = 2, tol = 2.454928e-15.
Warning: Rank deficient, rank = 2, tol = 2.454917e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'Mytilus.spp_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 2.9684e-12]
  delta: [-0.2500 0]
  Ed: [-1.4790e-12 2.9405e-12]
  r2m: -Inf
  Ao: [16.4824 258.6739]
  epsilon: [0 0]
  Eo: [0.1378 0.0649]
  r2p: 0.8290
```

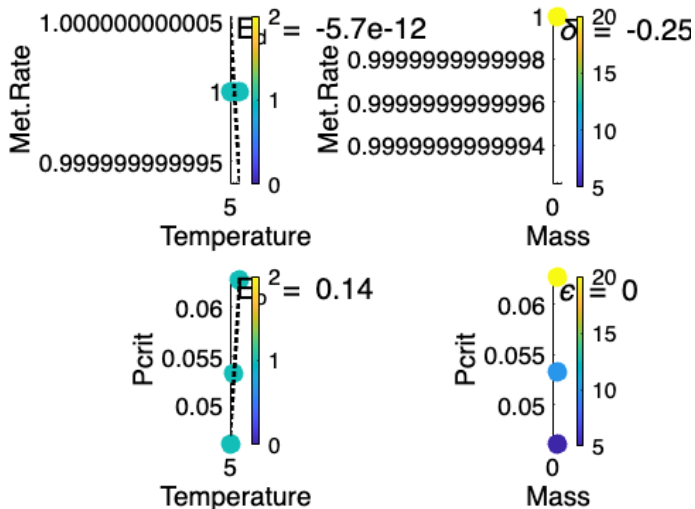
```
%0.gracilirostris
%Pcrit is atm
A=readtable('Deutsch2020PubData_Oplophorus_gracilirostris_Pcrit.atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 2.316904e-15.
Warning: Rank deficient, rank = 2, tol = 3.126841e-15.
Warning: Rank deficient, rank = 2, tol = 2.407750e-15.
Warning: Rank deficient, rank = 2, tol = 2.341619e-15.
Warning: Rank deficient, rank = 2, tol = 2.340791e-15.
Warning: Rank deficient, rank = 1, tol = 2.307555e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: Rank deficient, rank = 2, tol = 3.150550e-15.
Warning: Rank deficient, rank = 2, tol = 2.252857e-15.
Warning: Rank deficient, rank = 2, tol = 2.192844e-15.
Warning: Rank deficient, rank = 2, tol = 2.189334e-15.
Warning: Rank deficient, rank = 2, tol = 2.189341e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg '0.gracilirostris_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 3.5448e-12]
  delta: [-0.2500 0]
  Ed: [-5.6510e-12 0]
  r2m: -Inf
  Ao: [17.4218 977.4337]
  epsilon: [0 0]
```

```

Eo: [0.1381 0.0189]
r2p: 0.9815

```

```

%0.spinosus
%Pcrit is atm
A=readtable('Deutsch2020PubData_Oplophorus_spinosus_Pcrit.atm.csv');

```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```

[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);

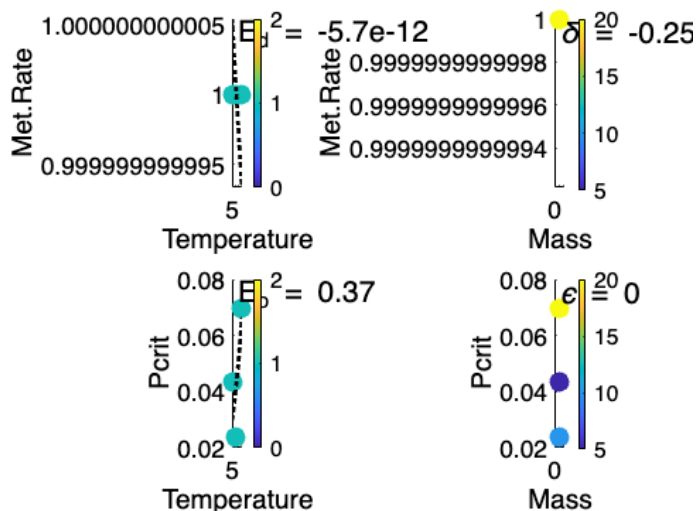
```

Warning: Rank deficient, rank = 2, tol = 2.316904e-15.
Warning: Rank deficient, rank = 2, tol = 3.126841e-15.
Warning: Rank deficient, rank = 2, tol = 2.407750e-15.
Warning: Rank deficient, rank = 2, tol = 2.341619e-15.
Warning: Rank deficient, rank = 2, tol = 2.340791e-15.
Warning: Rank deficient, rank = 1, tol = 2.307555e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: Rank deficient, rank = 2, tol = 3.150550e-15.
Warning: Rank deficient, rank = 2, tol = 2.189626e-15.
Warning: Rank deficient, rank = 2, tol = 2.141157e-15.
Warning: Rank deficient, rank = 2, tol = 2.145870e-15.
Warning: Rank deficient, rank = 2, tol = 2.147147e-15.
Warning: Rank deficient, rank = 2, tol = 2.147333e-15.
Warning: Rank deficient, rank = 2, tol = 2.147358e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```

print -djpeg '0.spinosus_MIparam_atm.jpg'

```



Par

Par = struct with fields:

```

alphaD: [1.0000 3.5448e-12]
delta: [-0.2500 0]
Ed: [-5.6510e-12 0]
r2m: -Inf
Ao: [19.4290 77.3755]
epsilon: [0 0]
Eo: [0.3738 0.3015]
r2p: 0.5945

```

```

%P.maximus
%Pcrit is atm
A=readtable('Deutsch2020PubData_Pecten_maximus_Pcrit.atm.csv');

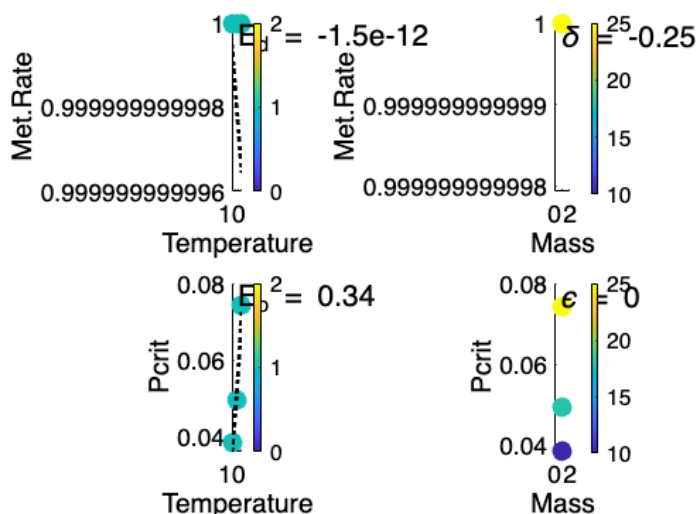
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 4.468923e-15.
Warning: Rank deficient, rank = 2, tol = 2.624281e-15.
Warning: Rank deficient, rank = 2, tol = 2.322380e-15.
Warning: Rank deficient, rank = 2, tol = 2.307774e-15.
Warning: Rank deficient, rank = 2, tol = 2.307556e-15.
Warning: Rank deficient, rank = 2, tol = 2.329442e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: Rank deficient, rank = 2, tol = 2.250037e-15.
Warning: Rank deficient, rank = 2, tol = 2.841797e-15.
Warning: Rank deficient, rank = 2, tol = 2.818111e-15.
Warning: Rank deficient, rank = 2, tol = 2.813069e-15.
Warning: Rank deficient, rank = 2, tol = 2.812860e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'P.maximus_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 2.9684e-12]
  delta: [-0.2500 0]
  Ed: [-1.4790e-12 2.9405e-12]
  r2m: -Inf
  Ao: [21.7463 326.7760]
  epsilon: [0 0]
  Eo: [0.3407 0.0619]
  r2p: 0.9715
```

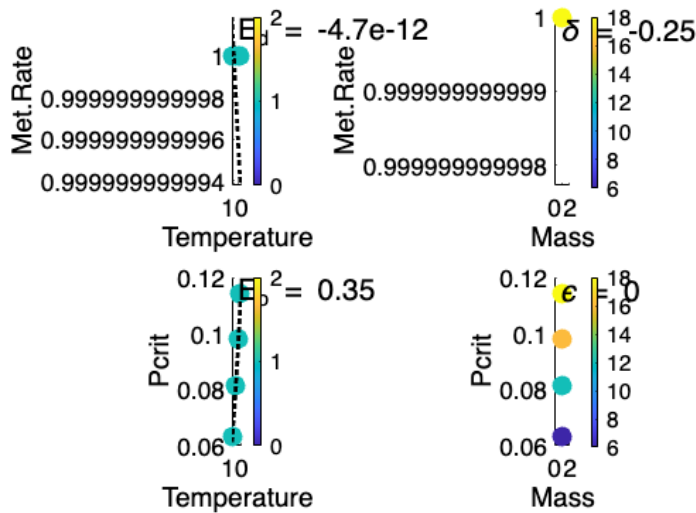
```
%S. salmar
%Pcrit is atm
A=readtable('Deutsch2020PubData_Salmo_salar_Pcrit.atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

```
Warning: Rank deficient, rank = 2, tol = 2.994924e-15.
Warning: Rank deficient, rank = 2, tol = 3.303825e-15.
Warning: Rank deficient, rank = 2, tol = 3.129527e-15.
Warning: Rank deficient, rank = 2, tol = 3.108904e-15.
Warning: Rank deficient, rank = 2, tol = 3.108626e-15.
Warning: Rank deficient, rank = 2, tol = 3.526170e-15.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the
model is insensitive to some of its parameters. That may be because those parameters are not
present in the model, or otherwise do not affect the predicted values. It may also be due to
numerical underflow in the model function, which can sometimes be avoided by choosing better
initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.
Warning: Rank deficient, rank = 2, tol = 3.777285e-15.
Warning: Rank deficient, rank = 2, tol = 3.004142e-15.
Warning: Rank deficient, rank = 2, tol = 2.946717e-15.
Warning: Rank deficient, rank = 2, tol = 2.943479e-15.
Warning: Rank deficient, rank = 2, tol = 2.943374e-15.
Warning: Rank deficient, rank = 2, tol = 2.943370e-15.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the
model is insensitive to some of its parameters. That may be because those parameters are not
present in the model, or otherwise do not affect the predicted values. It may also be due to
numerical underflow in the model function, which can sometimes be avoided by choosing better
initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.
```

```
print -djpeg 'S.salmar_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
    alphaD: [1.0000 3.4594e-12]
    delta: [-0.2500 0]
    Ed: [-4.7171e-12 3.0497e-12]
    r2m: -Inf
    Ao: [10.3731 537.4610]
    epsilon: [0 0]
    Eo: [0.3484 0.0353]
    r2p: 0.9830
```

```
%S. bisulcatus
```

```
%Pcrit is atm
```

```
A=readtable('Deutsch2020PubData_Sergestes_bisulcatus_Pcrit.atm.csv');
```

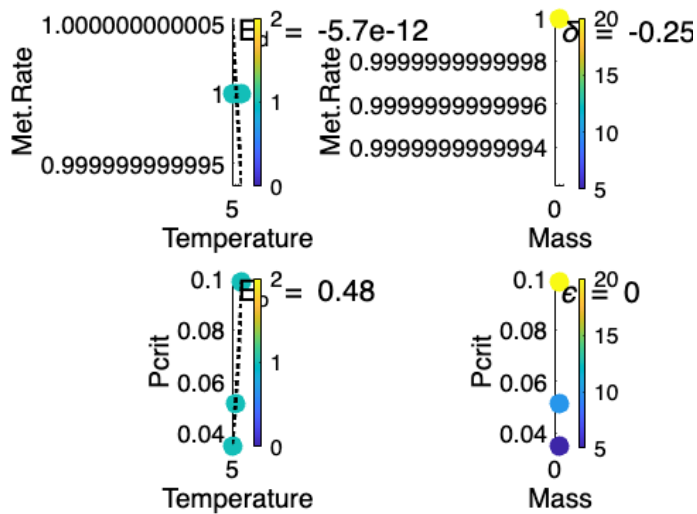
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

```
Warning: Rank deficient, rank = 2, tol = 2.316904e-15.
Warning: Rank deficient, rank = 2, tol = 3.126841e-15.
Warning: Rank deficient, rank = 2, tol = 2.407750e-15.
Warning: Rank deficient, rank = 2, tol = 2.341619e-15.
Warning: Rank deficient, rank = 2, tol = 2.340791e-15.
Warning: Rank deficient, rank = 1, tol = 2.307555e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will
not be able to compute confidence or prediction intervals, and you should use caution in making
predictions.
Warning: Rank deficient, rank = 2, tol = 3.150550e-15.
Warning: Rank deficient, rank = 2, tol = 2.143832e-15.
Warning: Rank deficient, rank = 2, tol = 2.173631e-15.
Warning: Rank deficient, rank = 2, tol = 2.183148e-15.
Warning: Rank deficient, rank = 2, tol = 2.183185e-15.
```

Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'S.bisulcatus_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 3.5448e-12]
  delta: [-0.2500 0]
  Ed: [-5.6510e-12 0]
  r2m: -Inf
  Ao: [14.0814 1.1173e+03]
  epsilon: [0 0]
  Eo: [0.4768 0.0158]
  r2p: 0.9991
```

```
%S. fulgens
%Pcrit is atm
A=readtable('Deutsch2020PubData_Sergestes_fulgens_Pcrit.atm.csv');
```

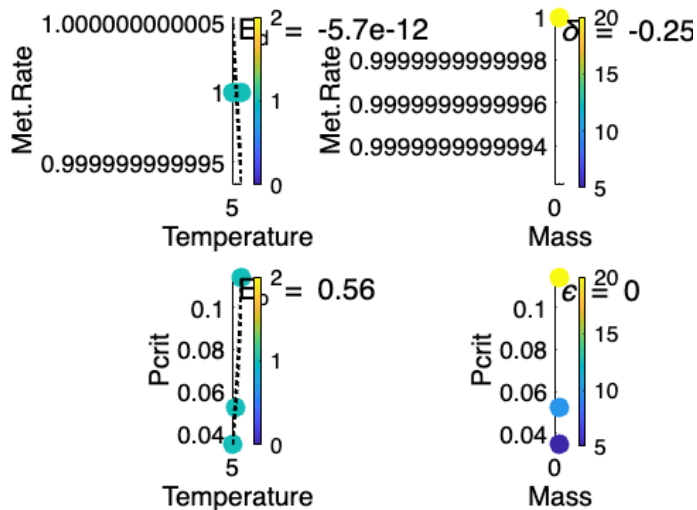
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 2.316904e-15.
Warning: Rank deficient, rank = 2, tol = 3.126841e-15.
Warning: Rank deficient, rank = 2, tol = 2.407750e-15.
Warning: Rank deficient, rank = 2, tol = 2.341619e-15.
Warning: Rank deficient, rank = 2, tol = 2.340791e-15.
Warning: Rank deficient, rank = 1, tol = 2.307555e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: Rank deficient, rank = 2, tol = 3.150550e-15.

Warning: Rank deficient, rank = 2, tol = 2.156046e-15.
Warning: Rank deficient, rank = 2, tol = 2.213396e-15.
Warning: Rank deficient, rank = 2, tol = 2.232346e-15.
Warning: Rank deficient, rank = 2, tol = 2.232627e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'S.fulgens_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 3.5448e-12]
  delta: [-0.2500 0]
  Ed: [-5.6510e-12 0]
  r2m: -Inf
  Ao: [12.8396 1.0556e+04]
  epsilon: [0 0]
  Eo: [0.5598 0.0016]
  r2p: 1.0000
```

```
%S. tenuiremis
%Pcrit is atm
A=readtable('Deutsch2020PubData_Sergestes_tenuiremis_Pcrit.atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

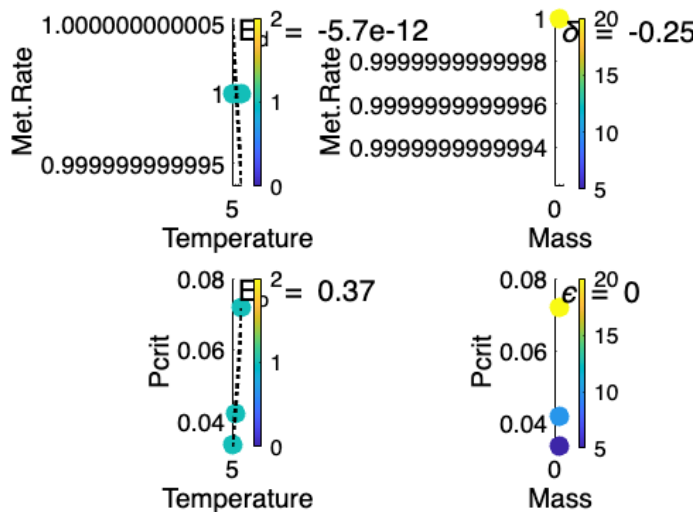
Warning: Rank deficient, rank = 2, tol = 2.316904e-15.
Warning: Rank deficient, rank = 2, tol = 3.126841e-15.
Warning: Rank deficient, rank = 2, tol = 2.407750e-15.
Warning: Rank deficient, rank = 2, tol = 2.341619e-15.
Warning: Rank deficient, rank = 2, tol = 2.340791e-15.
Warning: Rank deficient, rank = 1, tol = 2.307555e-15.

Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

Warning: Rank deficient, rank = 2, tol = 3.150550e-15.
Warning: Rank deficient, rank = 2, tol = 2.149086e-15.
Warning: Rank deficient, rank = 2, tol = 2.143575e-15.
Warning: Rank deficient, rank = 2, tol = 2.146058e-15.
Warning: Rank deficient, rank = 2, tol = 2.146100e-15.
Warning: Rank deficient, rank = 2, tol = 2.146099e-15.
Warning: Rank deficient, rank = 2, tol = 2.146100e-15.
Warning: Rank deficient, rank = 2, tol = 2.146110e-15.
Warning: Rank deficient, rank = 2, tol = 2.146206e-15.
Warning: Rank deficient, rank = 2, tol = 2.147172e-15.
Warning: Rank deficient, rank = 2, tol = 2.156803e-15.
Warning: Rank deficient, rank = 2, tol = 2.250848e-15.
Warning: Rank deficient, rank = 2, tol = 3.035043e-15.

Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'S.tenuiremis_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 3.5448e-12]
  delta: [-0.2500 0]
  Ed: [-5.6510e-12 0]
  r2m: -Inf
  Ao: [17.9666 1.3783e+03]
  epsilon: [0 0]
  Eo: [0.3672 0.0156]
  r2p: 0.9983
```

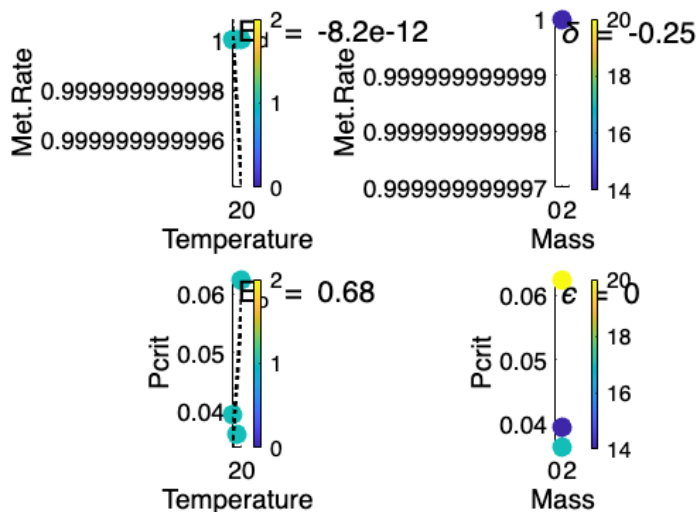
```
%S. robustus
%Pcrit is atm
A=readtable('Deutsch2020PubData_Sergia_robustus_Pcrit_atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 2.907446e-15.
Warning: Rank deficient, rank = 2, tol = 2.395089e-15.
Warning: Rank deficient, rank = 2, tol = 2.310412e-15.
Warning: Rank deficient, rank = 2, tol = 2.307571e-15.
Warning: Rank deficient, rank = 2, tol = 2.307556e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: Rank deficient, rank = 2, tol = 2.115344e-15.
Warning: Rank deficient, rank = 2, tol = 2.889762e-15.
Warning: Rank deficient, rank = 2, tol = 2.919242e-15.
Warning: Rank deficient, rank = 2, tol = 2.924697e-15.
Warning: Rank deficient, rank = 2, tol = 2.925259e-15.
Warning: Rank deficient, rank = 2, tol = 2.925316e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'S.robustus_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 4.8369e-12]
  delta: [-0.2500 0]
  Ed: [-8.1663e-12 1.1931e-11]
  r2m: -Inf
  Ao: [26.9152 120.1341]
  epsilon: [0 0]
  Eo: [0.6765 0.4264]
  r2p: 0.7121
```

```
%S. debilis
```

```
%Pcrit is atm
```

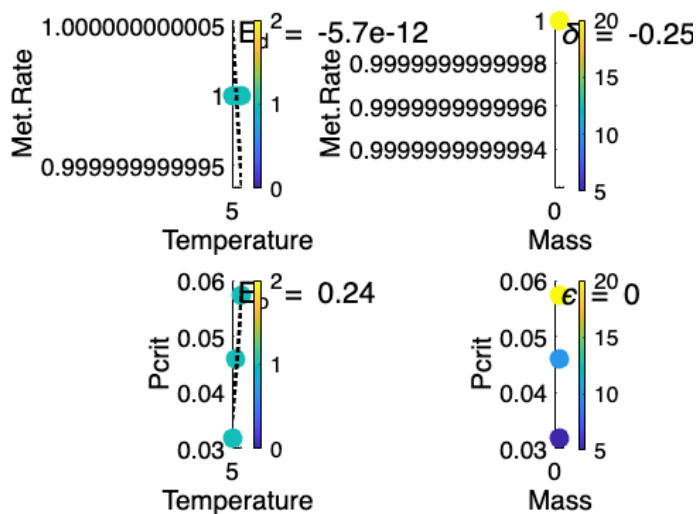
```
A=readtable('Deutsch2020PubData_Systellaspis_debilis_Pcrit.atm.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 2.316904e-15.
Warning: Rank deficient, rank = 2, tol = 3.126841e-15.
Warning: Rank deficient, rank = 2, tol = 2.407750e-15.
Warning: Rank deficient, rank = 2, tol = 2.341619e-15.
Warning: Rank deficient, rank = 2, tol = 2.340791e-15.
Warning: Rank deficient, rank = 1, tol = 2.307555e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.
Warning: Rank deficient, rank = 2, tol = 3.150550e-15.
Warning: Rank deficient, rank = 2, tol = 2.178038e-15.
Warning: Rank deficient, rank = 2, tol = 2.150520e-15.
Warning: Rank deficient, rank = 2, tol = 2.149036e-15.
Warning: Rank deficient, rank = 2, tol = 2.149069e-15.
Warning: The model is overparameterized, and model parameters are not identifiable. You will not be able to compute confidence or prediction intervals, and you should use caution in making predictions.

```
print -djpeg 'S.debilis_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 3.5448e-12]
  delta: [-0.2500 0]
  Ed: [-5.6510e-12 0]
  r2m: -Inf
  Ao: [20.1934 293.8471]
  epsilon: [0 0]
  Eo: [0.2367 0.0769]
  r2p: 0.9084
```

```
% Phys Lit Review Endress 2024, Duncan 2023, Boag 2018%
```

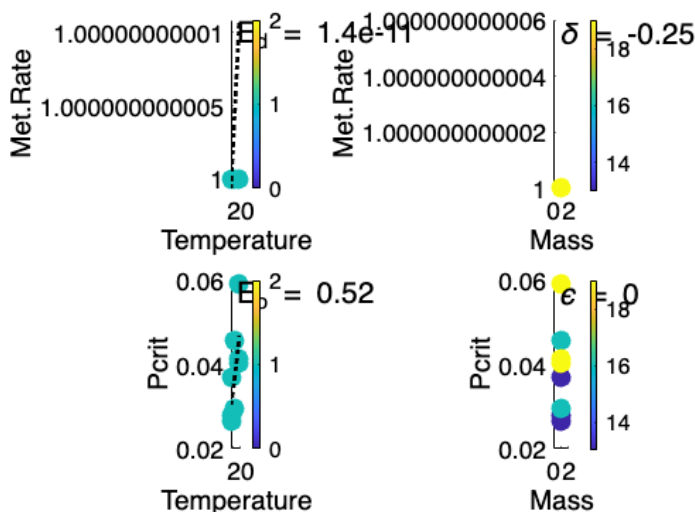
```
%d.opalescens
%Pcrit is atm
A=readtable('d.opalescens_formatlab_warmonly.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 8.002973e-15.
Warning: Rank deficient, rank = 2, tol = 6.981822e-15.
Warning: Rank deficient, rank = 2, tol = 6.910500e-15.
Warning: Rank deficient, rank = 2, tol = 6.908442e-15.
Warning: Rank deficient, rank = 2, tol = 6.908410e-15.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.
Warning: Rank deficient, rank = 2, tol = 6.711776e-15.
Warning: Rank deficient, rank = 2, tol = 7.739110e-15.
Warning: Rank deficient, rank = 2, tol = 7.677418e-15.
Warning: Rank deficient, rank = 2, tol = 7.673755e-15.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.

```
print -djpeg 'D.opalescens_MIparam_atm_warmonly.jpg'
```



```
Par
```

```
Par = struct with fields:
```

```

alphaD: [1.0000 3.6828e-12]
delta: [-0.2500 0]
Ed: [1.3745e-11 4.6914e-12]
r2m: -Inf
Ao: [28.4112 294.3262]
epsilon: [0 0]
Eo: [0.5184 0.2231]
r2p: 0.4894

```

```

%d.lineata
%Pcrit is atm
A=readtable('d.lineata_formatlab_warmonly.csv');

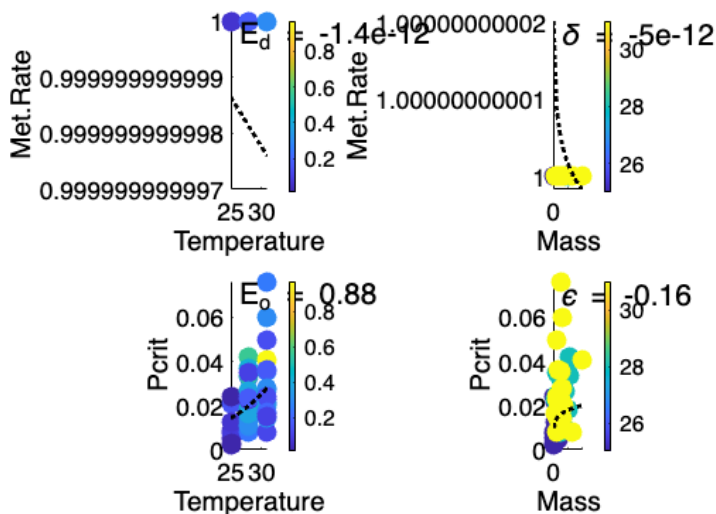
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 1.390380e-13.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.

```
print -djpeg 'd.lineata_MIparam_atm_wormonly.jpg'
```



Par

```

Par = struct with fields:
  alphaD: [1.0000 7.5929e-13]
  delta: [-4.9905e-12 4.9436e-13]
  Ed: [-1.3717e-12 0]
  r2m: -Inf
  Ao: [230.7310 333.5678]
  epsilon: [-0.1631 0.1142]

```

```

Eo: [0.8809 0.3559]
r2p: 0.2580

```

```

%l.pictus
%Pcrit is atm
A=readtable('l.pictus_formatlab_warmonly_removeNArow.csv');

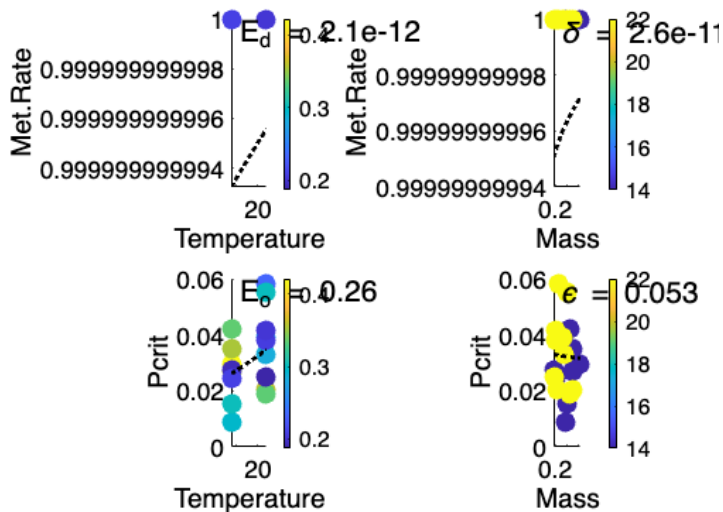
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```

[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'l.pictus_MIparam_atm_warmonly_removeNArow.jpg'

```



Par

```

Par = struct with fields:
  alphaD: [1.0000 3.1506e-12]
  delta: [2.6040e-11 1.0364e-11]
  Ed: [2.1361e-12 4.6576e-12]
  r2m: -Inf
  Ao: [36.8503 238.0837]
  epsilon: [0.0528 0.4310]
  Eo: [0.2616 0.2033]
  r2p: 0.1226

```

```

%n.vectensis
%Pcrit is atm
A=readtable('n.vectensis_formatlab_warmonly.csv');

```

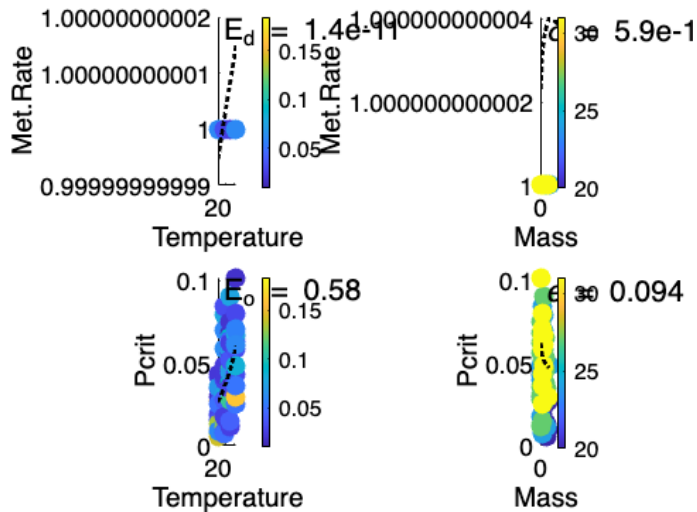
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```

[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);

```

```
print -djpeg 'N.vectensis_Miparam_atm_warmonly.jpg'
```



Par

```
Par = struct with fields:
    alphaD: [1.0000 2.8922e-12]
    delta: [5.8574e-13 1.4755e-12]
    Ed: [1.4263e-11 1.9110e-12]
    r2m: -Inf
    Ao: [56.5033 271.9557]
    epsilon: [0.0940 0.0794]
    Eo: [0.5779 0.1209]
    r2p: 0.2960
```

```
%c.latticeps
%Pcrit is atm
A=readtable('c.latticeps_formatlab.csv');
```

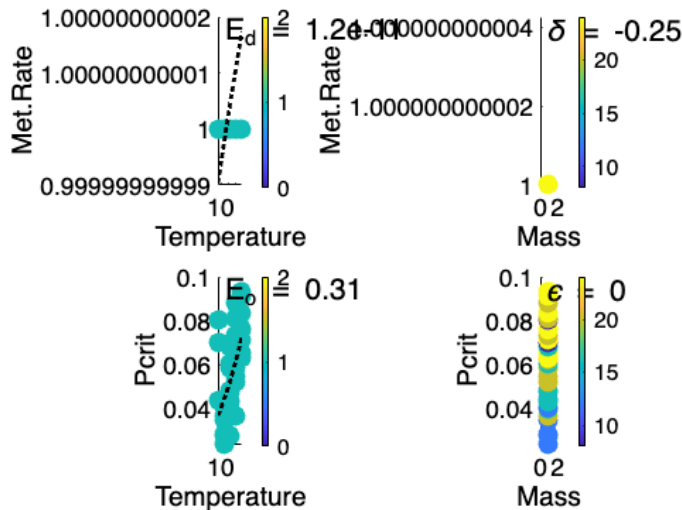
Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 8.783905e-14.
Warning: Rank deficient, rank = 2, tol = 6.295187e-14.
Warning: Rank deficient, rank = 2, tol = 5.849018e-14.
Warning: Rank deficient, rank = 2, tol = 5.824242e-14.
Warning: Rank deficient, rank = 2, tol = 5.824009e-14.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.
Warning: Rank deficient, rank = 2, tol = 5.499231e-14.
Warning: Rank deficient, rank = 2, tol = 6.425573e-14.
Warning: Rank deficient, rank = 2, tol = 6.712250e-14.
Warning: Rank deficient, rank = 2, tol = 6.777010e-14.

Warning: Rank deficient, rank = 2, tol = 6.783186e-14.
Warning: Rank deficient, rank = 2, tol = 6.783763e-14.
Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.

```
print -djpeg 'C.latticeps_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
  alphaD: [1.0000 2.0050e-12]
  delta: [-0.2500 0]
  Ed: [1.2113e-11 2.5673e-12]
  r2m: -Inf
  Ao: [20.0029 334.2600]
  epsilon: [0 0]
  Eo: [0.3084 0.0679]
  r2p: 0.3595
```

```
%m.brachydactyla
%Pcrit is atm
A=readtable('m.brachydactyla_formatlab.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
```

Warning: Rank deficient, rank = 2, tol = 4.520610e-14.
Warning: Rank deficient, rank = 2, tol = 2.582203e-14.
Warning: Rank deficient, rank = 2, tol = 1.997120e-14.
Warning: Rank deficient, rank = 2, tol = 1.943152e-14.
Warning: Rank deficient, rank = 2, tol = 1.943658e-14.
Warning: Rank deficient, rank = 2, tol = 4.852324e-14.

Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.

Warning: Rank deficient, rank = 2, tol = 1.271725e-14.

Warning: Rank deficient, rank = 2, tol = 3.230740e-14.

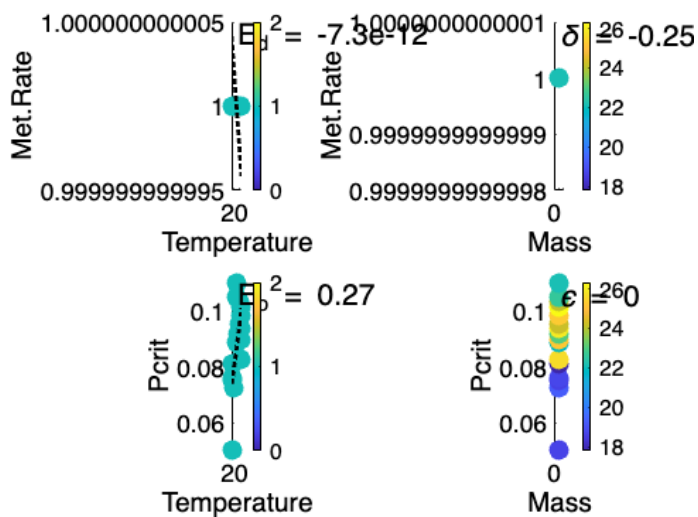
Warning: Rank deficient, rank = 2, tol = 2.424893e-14.

Warning: Rank deficient, rank = 2, tol = 2.427304e-14.

Warning: Rank deficient, rank = 2, tol = 2.427101e-14.

Warning: Some columns of the Jacobian are effectively zero at the solution, indicating that the model is insensitive to some of its parameters. That may be because those parameters are not present in the model, or otherwise do not affect the predicted values. It may also be due to numerical underflow in the model function, which can sometimes be avoided by choosing better initial parameter values, or by rescaling or recentering. Parameter estimates may be unreliable.

```
print -djpeg 'M.brachydactyla_MIparam_atm.jpg'
```



Par

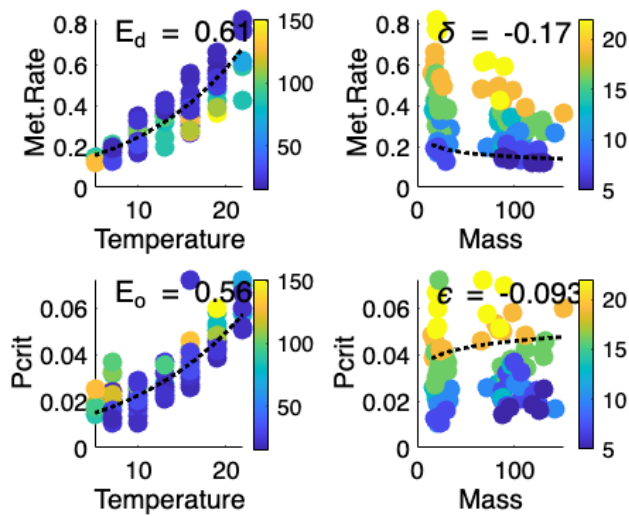
```
Par = struct with fields:
  alphaD: [1.0000 9.1169e-13]
  delta: [-0.2500 0]
  Ed: [-7.2733e-12 3.4403e-13]
  r2m: -Inf
  Ao: [15.0411 156.2230]
  epsilon: [0 0]
  Eo: [0.2743 0.0849]
  r2p: 0.4297
```

```
%S.purpuratus
%Pcrit is atm
A=readtable('s.purpuratus_formatlab.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.

Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'S.purpuratus_MIparam_atm.jpg'
```



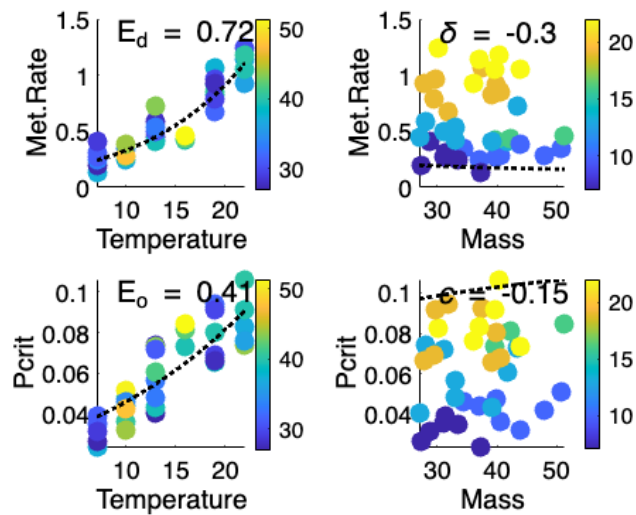
Par

```
Par = struct with fields:
  alphaD: [0.3800 0.0064]
  delta: [-0.1740 0.0209]
  Ed: [0.6128 0.0257]
  r2m: 0.8902
  Ao: [29.7436 1.2320e+03]
  epsilon: [-0.0932 0.0259]
  Eo: [0.5599 0.0324]
  r2p: 0.7784
```

```
%H. rufescens
%Pcrit is atm
A=readtable('h.rufescens_formatlab.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
[Par,mdlM,mdlP]=fit_MI_single(A.temp,A.afdw,A.pcrit,A.smr,mpjoint);
print -djpeg 'h.rufescens_MIparam_atm.jpg'
```



Par

```
Par = struct with fields:
    alphaD: [0.5539 0.0209]
    delta: [-0.2997 0.1710]
    Ed: [0.7238 0.0497]
    r2m: 0.8928
    Ao: [16.2512 495.4034]
    epsilon: [-0.1468 0.1832]
    Eo: [0.4082 0.0456]
    r2p: 0.7345
```

```
function [Par,mdlM,mdlP] = fit_MI_single(temp,mass,Pcrit,MR,mpjoint)

global kb T0 Tref

% fitting function: coef * PowerLaw(mass) * Arrhenius(Temp)
modelfun = @(b,x)b(1).*x(:,1).^b(2).*exp(-b(3)*x(:,2));

%% prepare variables for regression

% dependent variables
yp = Pcrit;
ym = MR;

% independent variables
mass(mass==0)=nan;
x1 = mass/nanmedian(mass);
x2 = (1./(temp+T0)-1./(Tref+T0)).*(1/kb); % Arrhenius exponent divided by E

% select/combine data
Ip=yp>0;
Im=ym>0;
if mpjoint
```

```

    Ij=Ip & Im;
else
    Ij=Ip | Im;
end

X=[x1(Ij)/median(x1(Ij)),x2(Ij)];
ym=ym(Ij);
yp=yp(Ij);

% fitting parameters
p0p=[nanmean(yp) 0 -0.4];
p0m=[nanmean(ym) -1/4 0.65];
tol=0.5;

% Metabolic rate fits
mdlM = fitnlm(X,ym,modelfun,p0m);
Par.alphaD=[mdlM.Coefficients.Estimate(1) mdlM.Coefficients.SE(1)];
Par.delta=[mdlM.Coefficients.Estimate(2) mdlM.Coefficients.SE(2)];
Par.Ed=[mdlM.Coefficients.Estimate(3) mdlM.Coefficients.SE(3)];
Par.r2m=mdlM.Rsquared.Ordinary;

% Metabolic Index fits
mdlP = fitnlm(X,yp,modelfun,p0p);
Par.Ao=1./[mdlP.Coefficients.Estimate(1) mdlP.Coefficients.SE(1)];
Par.epsilon=[-mdlP.Coefficients.Estimate(2) mdlP.Coefficients.SE(2)];
Par.Eo=[mdlP.Coefficients.Estimate(3) mdlP.Coefficients.SE(3)];
Par.r2p=mdlP.Rsquared.Ordinary;

%% plot results

% data ranges
rngT=linspace(min(temp),max(temp),10)';
rngTT=(1./(rngT+T0)-1./(Tref+T0)).*(1/kb);
rngB=linspace(min(mass),max(mass),10)';
X1=[rngTT*0+1,rngTT];
X2=[rngB,rngB*0+nanmedian(rngTT)];

% plots
figure(1); clf reset
set(gcf, 'Color', 'w') % <-- Ensures white background for the figure
fsz=15;
subplot(221)
scatter(temp(Ij),MR(Ij),temp(Ij)*0+100,mass(Ij),'filled');hold on;c =
colorbar; % create or get the colorbar
c.Color = 'k'; % set colorbar text (ticks/label) to black
plot(rngT,modelfun(mdlM.Coefficients.Estimate,X1),'k:','LineWidth',2);hold
on;
text(0.1,0.9,['E_d = '
num2str(Par.Ed(1),2)], 'FontSize',fsz,'units','normalized', 'Color', 'k')
xlabel('Temperature');ylabel('Met.Rate')

```

```

legend off
set(gca, 'Color', 'w', 'XColor', 'k', 'YColor', 'k')
subplot(222)
scatter(mass(Ij),MR(Ij),temp(Ij)*0+100,temp(Ij),'filled');hold on;c =
colorbar; % create or get the colorbar
c.Color = 'k'; % set colorbar text (ticks/label) to black
plot(rngB,modelfun mdlM.Coefficients.Estimate,X2),'k:','LineWidth',2);hold
on;
text(0.1,0.9,['\delta = '
num2str(Par.delta(1),2)],'FontSize',fsz,'units','normalized', 'Color', 'k')
xlabel('Mass');ylabel('Met.Rate')
set(gca, 'Color', 'w', 'XColor', 'k', 'YColor', 'k')
subplot(223)
scatter(temp(Ij),Pcrit(Ij),temp(Ij)*0+100,mass(Ij),'filled');hold on;c =
colorbar; % create or get the colorbar
c.Color = 'k'; % set colorbar text (ticks/label) to black
plot(rngT,modelfun mdlP.Coefficients.Estimate,X1),'k:','LineWidth',2);hold
on;
text(0.1,0.9,['E_o = '
num2str(Par.Eo(1),2)],'FontSize',fsz,'units','normalized', 'Color', 'k')
xlabel('Temperature');ylabel('Pcrit')
set(gca, 'Color', 'w', 'XColor', 'k', 'YColor', 'k')
subplot(224)
scatter(mass(Ij),Pcrit(Ij),temp(Ij)*0+100,temp(Ij),'filled');hold on;c =
colorbar; % create or get the colorbar
c.Color = 'k'; % set colorbar text (ticks/label) to black
plot(rngB,modelfun mdlP.Coefficients.Estimate,X2),'k:','LineWidth',2);hold
on;
text(0.1,0.9,['\epsilon = '
num2str(Par.epsilon(1),2)],'FontSize',fsz,'units','normalized', 'Color', 'k')
xlabel('Mass');ylabel('Pcrit')
set(gca, 'Color', 'w', 'XColor', 'k', 'YColor', 'k')
end

```